Meet EMMA: Your Emergency Medicine Mentoring Agent

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Abstract

Medical licensing examinations demand synthesis of clinical reasoning across dozens of specialties under high-stakes conditions. This paper presents EMMA (Emergency Medicine Mentoring Agent), a conversational study agent that grounds responses in passages from eighteen standard medical textbooks via a five-component machine learning pipeline: biomedical NER for clinical entity extraction, FAISS-based dense retrieval over 36,723 textbook chunks, TF-IDF + LinearSVC specialty classification, BERTopic clustering for fine-grained topic routing, and a collaborative filtering recommender. A six-run ablation grid varying embedding model, LLM, and RAG condition shows that textbook-grounded RAG delivers up to an 11 percentage-point accuracy gain on MedQA under the right embedding–LLM pairing, while degrading performance under others. These findings suggest that RAG benefit is highly sensitive to semantic alignment between the embedding space and retrieval corpus.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; *Machine learning*.

Keywords: retrieval-augmented generation, medical education, large language models, named entity recognition, topic modeling, collaborative filtering

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1 Introduction

Medical licensing examinations such as the United States Medical Licensing Examination (US-MLE) represent one of the most demanding knowledge challenges in professional education. Candidates must synthesize clinical reasoning across dozens of specialties under timed, high-stakes conditions. Traditional preparation relies on self-directed study, peer quizzing, and expensive tutoring — resources not equally accessible to all candidates.

This paper presents Emergency Medicine Mentoring Agent (EMMA), a conversational medical study agent built to close that gap. In explanation mode, students pose clinical questions in natural language and receive responses grounded in passages from eighteen standard medical textbooks. In quiz mode, EMMA presents authentic US-MLE-style questions, evaluates answers, and tracks per-specialty performance over time. A collaborative filtering recommender then steers students toward their weakest areas.

EMMA is built on a five-component Machine Learning (ML) pipeline: a SpaCy Named Entity Recognition (NER) model for clinical entity extraction, a Facebook AI Similarity Search (FAISS) vectorstore over 36,723 textbook chunks, a Term Frequency-Inverse Document Frequency (TF-IDF) + Linear Support Vector Classification (LinearSVC) specialty classifier, a BERTopic clustering model for fine-grained topic routing, and a Surprise-based Collaborative Filtering (CF) recommender. All components feed a locally-run Qwen3-4B-Thinking-2507 Large Language Model (LLM).

The central empirical question is whether textbook-grounded Retrieval-Augmented Generation (RAG) improves a small local LLM's accuracy on MedQA compared to a no-retrieval baseline. A six-run ablation grid shows the answer is condition-dependent: with the right embedding and LLM pairing, RAG delivers an +11 percentage points (pp) accuracy gain; without those conditions, it can hurt.

The remainder of this paper is organized as follows. Section **??** formalizes the problem. Sections **??**–**??** describe each pipeline component. Section **??** presents the RAG ablation study. Section **??** evaluates results. Section **??** analyzes failure modes. Section **??** concludes.

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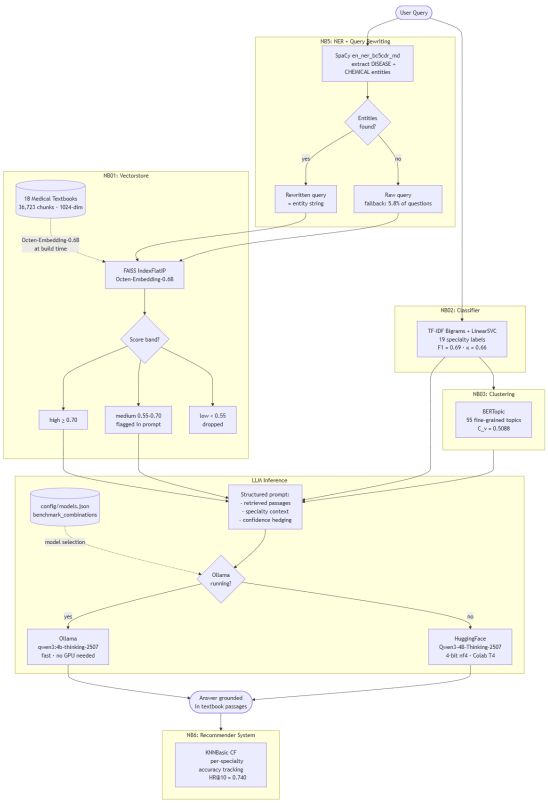

Figure 1. The EMMA system architecture.

open-source communities behind spaCy, FAISS, BERTopic, and the Surprise library.

References

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